Notes: Williamson (JFE, 2001)

Exchange Rate Exposure and Competition: Evidence from the Automotive Industry

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1 Big picture

- **Question.** Do real exchange-rate changes affect the value of multinational firms and global competitors, and does the degree of product-market competition determine the strength and sign of that exposure?
- **Main setting.** The paper studies U.S. and Japanese automobile manufacturers from January 1973 to December 1995. This is a strong setting because the auto industry has high export sales, direct cross-country competition, and large changes in industry structure over the sample.
- **Core idea.** Exchange-rate exposure is not determined only by foreign sales. It also depends on the elasticity of demand, the substitutability of foreign and domestic products, the location of production, and the identity of competitors in each market.
- **Why prior work found weak results.** Broad trade-weighted exchange-rate indices and cross-industry samples can mute exposure. A single exchange-rate index may mix currencies with offsetting effects, and heterogeneous industries have different product-market structures.
- **Empirical strategy.** Instead of using a trade-weighted exchange-rate index, the paper tests exposure to specific bilateral real exchange rates: yen-dollar, Deutschmark-dollar, and yen-Deutschmark.
- **Full-sample result for U.S. firms.** U.S. automakers have significant exposure to both the yen and the Deutschmark. The U.S. portfolio has a negative yen exposure and a positive Deutschmark exposure.
- **Full-sample result for Japanese firms.** Japanese automakers have exposure mainly to the dollar. Toyota, Nissan, and Honda have significant negative dollar exposures, consistent with substantial U.S. sales and competition in the U.S. market.
- **Competition result.** Market-share interactions show that exposure is tied to where firms compete. U.S. firms' yen exposure is driven by Japanese firms' U.S. market share, while Japanese firms' dollar exposure is tied to Japanese firms' U.S. market share.
- **Time-variation result.** Exposures change over time as the automobile industry moves from national competition to international and then global competition. The strongest exposures appear when foreign competition and sustained real exchange-rate movements are both important.
- **Operational hedging result.** For Japanese firms, U.S. sales increase dollar exposure, while U.S. production can reduce it by creating dollar-denominated costs. This is evidence for operational hedging through foreign production.
- **Economic takeaway.** Currency exposure is a strategic-product-market object, not just an accounting object. The same exchange-rate movement can help or hurt a firm depending on where it sells, where it produces, and which foreign competitors are active.

2 Data and measurement

2.1 Real exchange-rate data

- The paper focuses on **real** exchange rates rather than nominal rates because real exchange-rate changes capture deviations from purchasing power parity and therefore affect real competitive positions.
- The key currencies are the U.S. dollar, Japanese yen, German Deutschmark, and British pound. The empirical tables focus mainly on yen-dollar, Deutschmark-dollar, and yen-Deutschmark rates.
- Exchange-rate and price-level data come from Datastream International.
- Real exchange rates are constructed by deflating nominal exchange rates using monthly wholesale price indices.
- The paper uses monthly data because price indices are monthly and because monthly returns reduce nonsynchronous-trading problems.

- Exchange rates and stock-return indices are measured on the 15th day of each month to reduce possible end-of-month macro-announcement effects.

Interpretation: The key shock is a persistent real exchange-rate movement. If a nominal exchange-rate movement is fully offset by price-level movements, it should not change the real competitive position of firms.

2.2 Automotive firm sample

- The sample contains publicly traded automobile manufacturers headquartered in the United States and Japan.
- U.S. firms include General Motors, Ford, and Chrysler.
- Japanese firms include Toyota, Nissan, Honda, Isuzu, Mazda, Suzuki, and Mitsubishi.
- German firms are not included as main test firms because they do not face comparable competition for most of the sample period.
- Other countries with important auto producers are excluded because firms were often government-owned, protected, or not publicly traded for enough of the sample.
- Returns are measured in each firm's home currency and include dividends and other distributions.
- The main country portfolios are equally weighted portfolios of automobile firms from each country.

Interpretation: The industry selection is intentional. The automotive industry is globally competitive, has substantial export activity, and underwent major structural changes in foreign production and competition during the sample.

2.3 Market-share data

- Monthly sales and U.S. production data come from *Ward's Automotive Yearbook*.
- Japanese production and operations information comes from the Japanese Automotive Manufacturers Association.
- Foreign operations and affiliations of U.S. firms come from *Notable Corporate Chronologies*.
- Market shares are measured for U.S., Japanese, and German firms in the United States, Japan, and Germany.
- The market-share figures classify firms by the headquarters of the parent firm.
- Market-share measures are for automobiles and do not fully capture light trucks, which become increasingly important over time.

Interpretation: Market shares proxy for the competitive environment. If Japanese firms gain share in the United States, U.S. firms' exposure to yen movements should become stronger because yen movements affect the competitive position of Japanese producers.

2.4 Foreign sales and foreign production variables

- The paper studies Japanese firms' U.S. sales as a percentage of total global sales.
- It also studies Japanese firms' U.S. production as a percentage of U.S. sales.
- U.S. production is interpreted as a natural hedge: it creates dollar-denominated costs that offset dollar-denominated revenues.

- The foreign-production analysis begins when Japanese firms start meaningful U.S. production.

Interpretation: Foreign sales increase exposure because more cash flows are denominated in the foreign market. Foreign production can reduce exposure because revenues and costs become more closely matched in the same currency.

3 Models and empirical specifications

3.1 Firm value and exchange-rate exposure

The paper begins with the value effect of exchange-rate changes:

$$\frac{dV}{dS} = \frac{1-q}{\rho}\frac{d\pi}{dS}, \tag{1}$$

where V is firm value, S is the spot exchange rate, q is the tax rate, ρ is the discount rate, and π is firm profit.

Interpretation: If taxes and discount rates are treated as fixed, exchange-rate exposure comes from how profits respond to exchange-rate movements. The central empirical issue is therefore not exchange rates per se, but how exchange rates move cash flows through sales, costs, and competition.

3.2 Baseline exposure regression

The baseline regression is

$$r_{j,t} = \alpha_j + \beta_j^m R_{m,t} + \sum_{k=1}^{n} \beta_{j,k}^e \Delta S_{k,t} + \varepsilon_{j,t}, \tag{2}$$

where $r_{j,t}$ is the real return of firm or country portfolio j, $R_{m,t}$ is the real return on the country-specific market portfolio, $\Delta S_{k,t}$ is the real exchange-rate change for currency k, and $\beta_{j,k}^e$ is the exchange-rate exposure.

Interpretation: The market factor removes ordinary stock-market comovement. The exchange-rate coefficients are therefore interpreted as residual firm-value sensitivity to currency movements after controlling for domestic market risk.

3.3 SUR estimation

At the firm level, the paper estimates the exposure equations using seemingly unrelated regressions (SUR). The reason is that residuals across firms in the same country can be correlated.

Interpretation: If residuals are cross-correlated, SUR improves efficiency and allows formal tests of whether exposure coefficients differ across firms. If residuals are not correlated, SUR reduces to ordinary least squares.

3.4 Market-share interaction specification

To test whether competition determines exposure, the paper estimates interactions between exchange-rate changes and market shares. For the U.S. portfolio, the representative specification is

$$r_t = \alpha + \beta^m R_{m,t} + \gamma_1 \Delta S_{¥,t} MS_{JP,US} + \gamma_2 \Delta S_{¥,t} MS_{US,JP} \tag{3}$$

$$+ \delta_1 \Delta S_{DM,t} MS_{GR,US} + \delta_2 \Delta S_{DM,t} MS_{US,GR} + \varepsilon_t, \tag{4}$$

where $MS_{A,B}$ is the market share of firms from country A in country B.

Interpretation: This regression asks whether a firm's exposure to a currency is stronger when firms from that currency area are stronger competitors in a key market. It converts the qualitative theory of competition into a time-varying empirical measure.

3.5 Time-varying exposure specification

To study changing industry structure, the paper splits the sample into three periods: 1973–1980, 1981–1988, and 1989–1995. The regression is

$$r_{j,t} = \beta_j^m R_{m,t} + \sum_{k=1}^{n} \beta_{j,k}^{e,p} \Delta S_{k,t} PDUM_{p,t} + \varepsilon_{j,t}, \tag{5}$$

where $PDUM_{p,t}$ is a dummy for period p.

Interpretation: The coefficients are allowed to differ across subperiods. This is central because the paper argues that exposure changes as competition and foreign production change.

3.6 Foreign sales and production specification

For Japanese firms, the paper estimates

$$r_t = \alpha + \beta^m R_{m,t} + \gamma_1 \Delta S_{¥,t} TS_{JP} + \lambda_2 \Delta S_{¥,t} Pr_{JP,US} + \varepsilon_t, \tag{6}$$

where TS_{JP} is U.S. sales as a percentage of total global sales for Japanese firms and $Pr_{JP,US}$ is U.S. production as a percentage of U.S. sales.

Interpretation: The coefficient on U.S. sales captures the foreign-revenue channel. The coefficient on U.S. production captures operational hedging: production in the destination market creates local-currency costs and can reduce net currency exposure.

4 Section-by-section notes

4.1 Section 1: Introduction

- The introduction motivates the paper with the puzzle that exchange rates are volatile, but many empirical studies find weak firm-level exchange-rate exposure.
- Williamson argues that weak results may come from misspecified empirical designs rather than from the absence of exposure.
- Existing broad-sample tests may mix firms with different currencies, markets, hedging policies, and competitive structures.
- The paper therefore focuses on one industry where exposure should be strong: automobiles.
- The central prediction is that firms with high foreign sales and high exposure to foreign competitors should display meaningful exchange-rate exposure.
- The introduction previews three main findings: significant exposures, time variation in exposure as industry competition changes, and evidence that foreign sales increase exposure while foreign production reduces it.

Interpretation: The introduction reframes exchange-rate exposure as a product-market competition problem. The main question is not simply whether firms sell abroad, but whether exchange-rate movements shift competitive advantage between firms.

4.2 Section 2: Review of exchange-rate exposure literature

- The literature defines exposure as the sensitivity of asset or firm value to real exchange-rate changes, following Adler and Dumas.
- Jorion finds cross-sectional and time variation in exchange-rate exposure and emphasizes foreign sales ratios.
- Bodnar and Gentry show that some industries have significant exposure and that industry characteristics matter.
- Bartov and Bodnar argue that lagged responses and large currency adjustments can reveal stronger exposure.
- He and Ng find significant exposure for Japanese multinationals, especially in transport-related sectors.
- Shapiro and Marston provide the theoretical basis for linking exposure to exports, market competition, and demand substitutability.
- Williamson's contribution is to combine currency-specific exposure tests with an explicit role for industry competition.

Interpretation: The paper's design responds directly to limitations in prior work: broad exchange-rate indices, cross-industry heterogeneity, and insufficient attention to competition can all hide exposure.

4.3 Section 3: Exchange-rate exposure and firm value

- A pure exporter with home-currency costs and foreign-currency sales is exposed because exchange-rate changes alter home-currency cash flows.
- The sensitivity depends on demand elasticity. If demand is inelastic, the exporter can pass through currency changes into local prices.
- Competition reduces pass-through. When local competitors are present, exporters cannot raise prices as easily after a foreign-currency depreciation.
- Foreign production can reduce exposure by creating local-currency costs that offset local-currency revenues.
- In global competition, exposure can have counterintuitive signs because firms compete in third markets. A currency depreciation can help a firm if it improves the competitiveness of its local operations relative to another foreign rival.
- The theoretical channel is strongest under Cournot-type competition, where profits depend on own demand elasticity, cross elasticities, marginal costs, and rivals' actions.

Interpretation: This section explains why the sign of exposure is not mechanical. A U.S. firm's exposure to the Deutschmark, for example, depends on whether it has European production and whether Japanese competitors lack similar local costs.

4.4 Section 4: Sample selection and exposure measurement

- The paper uses real exchange rates because real changes affect competitiveness.

- The sample covers monthly observations from January 1973 to December 1995.
- U.S. and Japanese automotive firms are selected because they dominate contestable global auto cash flows and directly compete in major markets.
- German firms are excluded from the main portfolio tests because they face less comparable competition over the full period.
- Returns are home-currency returns; country-specific stock-market returns are used as market controls.
- The baseline exposure regression includes bilateral real exchange-rate changes rather than a trade-weighted index.

Interpretation: The data design is meant to match the theory: currency-specific real shocks in a single globally competitive industry.

4.5 Section 5: Full-sample industry and firm exposure results

- Table 1 first shows that exchange-rate changes are correlated. This motivates currency-specific interpretation and explains why multiple currencies in one regression can reduce precision.
- Table 2 shows that the U.S. portfolio has significant yen and Deutschmark exposure.
- The U.S. yen exposure is negative, meaning U.S. firms lose when the yen depreciates relative to the dollar. This is consistent with Japanese competitors gaining cost advantage in the U.S. market.
- The U.S. Deutschmark exposure is positive. Williamson interprets this as a European competition effect: U.S. firms with European operations can benefit from Deutschmark depreciation if it improves their competitive position relative to Japanese exporters.
- Table 3 shows that Japanese firms' significant exposure is concentrated in the dollar, especially for Toyota, Nissan, and Honda.
- The Deutschmark exposure of Japanese firms is generally weaker in the full sample.

Interpretation: The full-sample tables establish that exposure exists. The signs already suggest that product-market competition is central to interpretation.

4.6 Section 5.2: Determinants of exposure

- The paper uses market shares in the United States, Japan, and Germany to connect exposure to competition.
- Figures 2–4 document the evolution of market shares.
- Japanese firms gain share in the U.S. market, while U.S. firms remain essentially absent from Japan.
- In Germany, Japanese firms gain share at the expense of U.S. firms, while German firms retain a large share.
- Table 4 interacts exchange-rate movements with market shares.
- U.S. firms' yen exposure is driven primarily by Japanese firms' U.S. market share.
- Japanese firms' dollar exposure is also linked to the U.S. market, where Japanese firms have substantial sales.

Interpretation: Market-share interactions show that currency exposure is not just about where a firm is headquartered. It is about where it competes and who it competes against.

4.7 Section 6: Time variation in exchange-rate exposures

- The paper divides the sample into three periods because industry competition and exchange-rate regimes change over time.
- The first period, 1973–1980, follows the breakdown of Bretton Woods and includes volatile currencies and the oil crisis. Competition is still mostly national.
- The second period, 1981–1988, includes a dollar appreciation and later depreciation around the Plaza Accord. Japanese competition becomes much more important in North America.
- The third period, 1989–1995, features more global competition, Japanese foreign production, and Japanese entry into Europe.
- Table 5 shows that U.S. exposures are strongest in the later periods, particularly to the Deutschmark in the third period.
- Table 6 shows that Japanese dollar exposure is strong in the second period but weakened in the third period as U.S. production expands and provides operational hedging.

Interpretation: The time-variation results are the paper's most direct evidence that exposure evolves with the competitive environment.

4.8 Section 7: Foreign sales, foreign operations, and exposure

- The paper emphasizes net exposure: exposure after the firm has chosen sales locations, production locations, and financial or operational hedges.
- Japanese firms expand U.S. sales and then expand U.S. production.
- U.S. production by Japanese firms grows from near zero in 1981 to around 30% of U.S. sales by 1989 and about 60% by 1995.
- Table 7 documents the operational expansion, including Honda's U.S. manufacturing, Nissan's U.S. and European production, Toyota's U.S. and Canadian production, and various alliances.
- Table 8 shows that U.S. sales raise Japanese firms' dollar exposure, while U.S. production tends to mitigate exposure.
- The Honda result is partly counterintuitive and may reflect financial hedging or firm-specific profit exposure.

Interpretation: The operational-hedging evidence is important because observed stock-return exposure is net exposure. Low exposure does not necessarily mean exchange rates do not matter; it can mean the firm has changed operations to offset risk.

4.9 Section 7.1: Yen exposure and the auto industry

- Williamson argues that the common underlying currency exposure in the U.S.-Japan automotive rivalry is yen exposure.
- U.S. firms are exposed to the yen through Japanese competition in the U.S. market.
- Japanese firms are exposed to the dollar through U.S. sales and U.S. competition.
- When the yen trade-weighted exchange rate is used, U.S. and Japanese portfolio coefficients have similar magnitudes and opposite signs.

Interpretation: The result reinforces the idea that exposure is a competitive relation between country-industry portfolios, not merely a translation of foreign revenues.

4.10 Section 8: Conclusion

- The paper concludes that exchange-rate exposure exists and is economically meaningful in the globally competitive automobile industry.
- Exposure varies across countries, across firms, and across time.
- Industry competition, foreign market shares, foreign sales, and foreign production all help explain this variation.
- The paper recommends that future exchange-rate exposure research model industry structure explicitly rather than relying only on broad trade-weighted exchange-rate indices.

Interpretation: The main lesson is methodological and economic: currency exposure should be studied in settings where the relevant currencies, competitors, and product markets are observable.

5 Identification and empirical design

5.1 Why the automotive industry is useful

- The auto industry is internationally competitive and currency-sensitive.
- The same firms compete in the same major markets, making cross-country comparisons cleaner than in broad samples.
- The industry changes substantially over 1973–1995, allowing the paper to examine exposure dynamics.
- Market shares and production locations are observable enough to test the mechanisms.

Interpretation: The paper does not claim random assignment of exchange-rate shocks. Its credibility comes from matching currency exposures to specific competitive channels in a well-defined industry.

5.2 Why bilateral currencies matter

- Trade-weighted indices can mask offsetting currency effects.
- Theory predicts currency-specific exposure: U.S. firms' yen exposure should differ from Deutschmark exposure.
- The automotive rivalry is centered around specific currency-country pairs.

Interpretation: Using bilateral real exchange rates is essential because the sign and magnitude of exposure depend on the specific competitor and market.

5.3 Why market controls matter

- The regressions include a country-specific stock-market return to control for general market movements.
- This avoids interpreting aggregate market shocks as currency exposure.
- Adjusted R^2 values are high enough to indicate that market controls capture a large part of monthly return variation.

Interpretation: The exchange-rate coefficients are residual exposures after broad domestic market risk has been removed.

5.4 Main limitations

- The paper observes net exposure, not all underlying hedging contracts.
- Derivative hedging data are unavailable for much of the sample.
- Market-share measures exclude light trucks, which become more important over time.
- Exchange rates are correlated, especially European currencies, which can complicate interpretation.
- German firms are excluded from main firm-level tests, limiting symmetry across countries.
- The design is industry-specific; the advantage is clarity, but the estimates are not automatically generalizable to all industries.

Interpretation: These limitations do not overturn the central evidence, but they matter when interpreting the coefficients as net effects of currencies, competition, operations, and hedging choices.

6 Tables and figures

6.1 Figure 1 and Table 1: currency variation and correlation structure

Read:

- Figure 1 plots real exchange-rate levels from 1973 to 1995.
- The dollar weakens after Bretton Woods, strengthens in the early 1980s, and weakens again after the Plaza Accord.
- The yen-to-Deutschmark rate is much more stable around its initial level than the dollar-related rates.
- Table 1 shows that the yen-dollar and Deutschmark-dollar real rates are positively correlated at 0.591.
- The yen-Deutschmark and Deutschmark-dollar correlation is negative at -0.519.
- Return correlations show that U.S. and Japanese auto portfolios are correlated but less than some broad market correlations.

Economic message: Currency movements are large and persistent, so they can affect real competitive positions. Correlation among exchange rates also explains why currency-specific tests are necessary.

(a) Figure 1: real exchange-rate levels

industry to that of the overall market in the respective countries.

Table 1
Exchange rate and returns correlation coefficients, 1973–1995

Panel A shows the correlation coefficients among the currencies of interest for the automotive industry for the full sample from 1/15/73 to 12/15/95. The currencies covered are the Deutschmark (DM), the U.S. Dollar ($), and the Japanese Yen (¥). Panel B is the correlation matrix for the monthly returns of the equally weighted portfolio of automotive firms for each country and the Datastream International value-weighted portfolio for each country from 1/15/73 to 12/15/95.

	¥/$	DM/$	
Panel A: Exchange rates			
DM/$	0.591		
¥/DM	0.382	– 0.519	
	U.S. Auto	Japan Auto	U.S. market
Panel B: Returns			
Japan Auto	0.224		
U.S. market	0.606	0.266	
Japan market	0.267	0.665	0.337

(b) Table 1: exchange-rate and return correlations

6.2 Tables 2 and 3: full-sample exposure for U.S. and Japanese firms

Read:

- Table 2 shows that the U.S. portfolio has yen exposure of -0.3413 with a 10% significance level and Deutschmark exposure of 0.3544 with a 5% significance level.
- GM is the main driver: GM has yen exposure of -0.3335 and Deutschmark exposure of 0.5170, both significant.
- Ford and Chrysler have the expected signs but weaker statistical significance.
- Table 3 shows that the Japanese portfolio's dollar exposure is negative but not significant in the full sample.
- At the firm level, Toyota, Nissan, and Honda have significant negative dollar exposures.
- Japanese firms' Deutschmark exposures are generally negative but statistically weak.

Economic message: The full-sample evidence establishes that auto firms have currency exposure, but firm-level differences matter. U.S. exposure is strongest for GM, while Japanese exposure is concentrated among firms with large U.S. sales.

Table 2
U.S. portfolio and firm-specific exchange rate exposure

Regressions of the automotive industry for the total sample are shown using the following model:

$$r_t = \alpha + \beta^m R_{mt} + \sum_{i=1}^{n} \beta^e \Delta S_t + \varepsilon_t,$$

where β^m is the market risk, R_{mt} is the return on the country-specific market portfolio, β^e is the currency exposure of the portfolio, ΔS_t is the change in real exchange rates, r_t is the monthly return, and ε_t is the error term. Data is taken from Datastream International and represents 276 observations covering 1973–1995. For the results, \$ is the U.S. dollar, DM is the Deutschmark, and ¥ is the Japanese yen. Heteroscedastic-consistent t-statistics are in parentheses, and ***,**, * denotes 1%, 5%, and 10% significance levels, respectively. The adjusted R^2 shown is from the ordinary least squares regression. Firm variation is an F-test of the difference in exposure across firms.

Firm	Intercept	Country-specific market risk	¥/\$	DM/\$	Adjusted R^2 (%)
U.S. portfolio	0.0028 (0.741)	1.0588 (13.697)***	− 0.3413 (− 1.874)*	0.3544 (2.098)**	37.3
GM	0.0012 (0.312)	0.9197 (11.879)***	− 0.3335 (− 2.003)**	0.5170 (2.959)***	32.4
Ford	0.0044 (1.072)	1.0048 (9.915)***	− 0.2422 (− 1.233)	0.1400 (0.789)	31.2
Chrysler	0.0029 (0.464)	1.2495 (9.568)***	− 0.4472 (− 1.436)	0.4067 (1.563)	23.0
Firm variation [F-test]			[0.426]	[3.572]**	

(a) Table 2: U.S. portfolio and firm-specific exposure

Table 3
Japan portfolio and firm-specific exchange rate exposure

Regressions of the automotive industry for the total sample are shown using the following model:

$$r_t = \alpha + \beta^m R_{mt} + \sum_{i=1}^{n} \beta^e \Delta S_t + \varepsilon_t,$$

where, β^m is the market risk, R_{mt} is the return on the country-specific market portfolio, β^e is the currency exposure of the portfolio, ΔS_t is the change in real exchange rates, r_t is the monthly return, and ε_t is the error term. Data is taken from Datastream International and represents 276 observations covering 1973–1995. For the results, \$ is the U.S. dollar, DM is the Deutschmark, and ¥ is the Japanese yen. Heteroscedastic-consistent t-statistics are in parentheses, and ***,**, * denotes 1%, 5%, and 10% significance levels, respectively. The adjusted R^2 shown is from the ordinary least squares regression. Firm variation is an F-test of the difference in exposure across firms.

Firm	Intercept	Country-specific market risk	\$/¥	DM/¥	Adjusted R^2 (%)
Japanese portfolio	0.0052 (1.602)	1.0113 (12.945)***	− 0.2196 (− 1.423)	− 0.1065 (− 0.700)	44.7
Toyota	0.0096 (1.880)*	0.8468 (7.343)***	− 0.4068 (− 2.363)**	0.0422 (0.213)	19.1
Nissan	0.0036 (0.911)	0.8696 (9.767)***	− 0.2487 (− 1.736)*	− 0.0621 (− 0.351)	28.7
Honda	0.0068 (1.266)	0.9119 (7.184)***	− 0.5234 (− 1.817)*	− 0.3286 (− 1.242)	21.1
Isuzu	0.0030 (0.466)	1.4249 (7.561)***	0.0601 (0.224)	0.2155 (0.770)	29.0
Mazda	0.0006 (0.112)	1.0104 (8.616)***	− 0.2273 (− 1.096)	− 0.2421 (− 1.122)	24.6
Suzuki	0.0086 (1.462)	0.8589 (7.631)***	− 0.2404 (− 0.689)	− 0.0564 (− 0.197)	14.8
Mitsubishi	0.0045 (0.981)	1.1389 (10.170)***	0.0527 (0.279)	− 0.3174 (− 1.632)	34.4
Firm variation [F-test]			[1.103]	[1.174]	

(b) Table 3: Japanese portfolio and firm-specific exposure

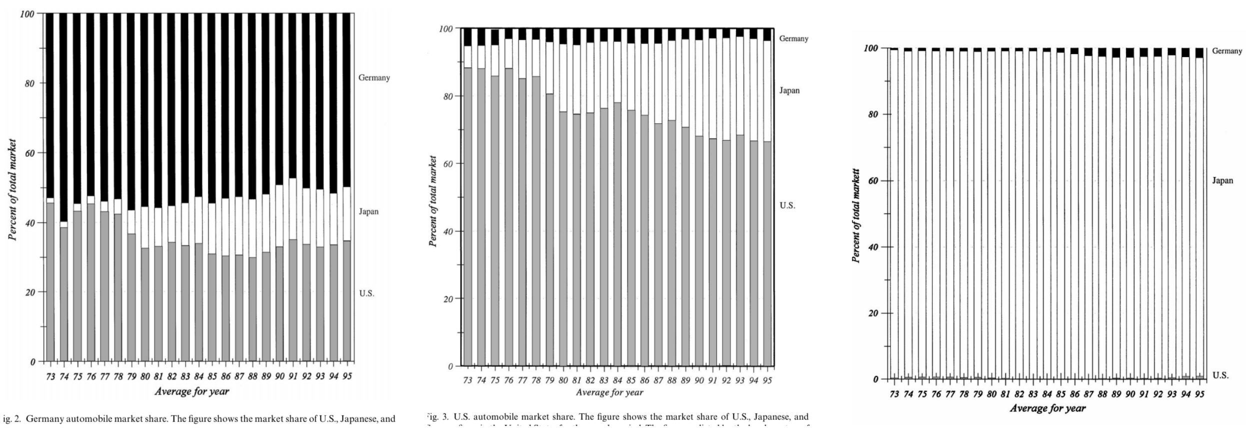
6.3 Figures 2–4: market shares in Germany, the United States, and Japan

Read:

- Figure 2 shows Germany's market: German firms dominate, U.S. firms lose share, and Japanese firms gain share over the sample.
- Figure 3 shows the U.S. market: U.S. firms remain dominant but steadily lose market share to Japanese firms.

- Figure 4 shows the Japanese market: Japanese firms dominate almost completely; U.S. and German firms have tiny shares.
- The figures show asymmetry. Japanese firms penetrate the U.S. market, but U.S. firms barely penetrate Japan.
- The figures also help explain why U.S. firms have yen exposure and why Japanese firms' dollar exposure is important.

Economic message: Market shares are the bridge between currency movements and competitive pressure. Exchange-rate exposure becomes stronger where foreign competitors have meaningful market presence.



(a) Figure 2: Germany market share **(b)** Figure 3: U.S. market share **(c)** Figure 4: Japan market share

6.4 Table 4: market-share and exchange-rate interactions

Read:

- The key U.S. result is that the interaction between Japanese firms' U.S. market share and yen-dollar movements is -4.0122 and significant at the 5% level.
- This means U.S. yen exposure is stronger when Japanese firms have a larger U.S. market share.
- The Japanese-firm result shows that the interaction between Japanese firms' U.S. market share and dollar-yen movements is -2.4210 and significant at the 5% level.
- The interaction between Japanese market share for German firms and Deutschmark-yen movements is negative and significant at the 10% level.
- The table supports the idea that exposure is generated by the product-market location of competition, not just by headquarters country.

Economic message: The table is the main mechanism test. It connects exchange-rate exposure to actual competition in specific markets.

6.5 Tables 5 and 6: time variation in exposure

Read:

- Table 5 shows that U.S. exposures are weak in 1973–1980, become more visible in 1981–1988, and are strongest in 1989–1995.

Table 4
Market share and exchange rate interaction

The table represents results from the following representative regression for each portfolio:

$$r_t = \alpha + \beta^m R_{mt} + \gamma_1 \Delta S_{¥,t}\ MS_{JP,US} + \gamma_2 \Delta S_{¥,t}\ MS_{US,JP} + \delta_1 \Delta S_{DM,t}\ MS_{GR,US} + \delta_2 \Delta S_{DM,t}\ MS_{US,GR} + \varepsilon_t,$$

where, β^m is the market risk, R_{mt} is the return on the country's market, γ_i and δ_i are the exposure of the interaction between the exchange rate and portfolio market share, ΔS_t is the rate of change in real exchange rates, $MS_{A,B}$ is the market share of portfolio A in country B, r_t is the monthly return, and ε_t is the error. Heteroscedastic-consistent t-statistics are in parentheses, and ***,**, * denotes 1%, 5%, and 10% significance levels, respectively.

	U.S. firms	Japanese firms
Intercept	0.0074 (1.460)	0.0057 (1.332)
Country-specific market risk	1.0524 (10.852)***	0.9213 (11.036)***
Interaction – U.S. market share for Japanese firms and ¥/$	− 4.0122 (− 2.152)**	
Interaction – Japanese market share for U.S. firms and ¥/$	55.1543 (1.284)	
Interaction – German market share for U.S. firms and DM/$	3.1938 (1.087)	
Interaction – U.S. market share for German firms and DM/$	− 13.0932 (− 0.468)	
Interaction – U.S. market share for Japanese firms and $/¥		− 2.4210 (− 2.114)**
Interaction – German market share for Japanese firms and DM/¥		8.7207 (1.563)
Interaction – Japanese market share for U.S. firms and $/¥		38.7183 (1.125)
Interaction – Japanese market share for German firms and DM/¥		− 65.2578 (− 1.708)*

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ariation of U.S. portfolio and firm-specific exchange rate exposure

egressions of the automotive industry for the total sample and three subperiods, using the following model:

$$r_t = \beta^m R_{mt} + \sum_{i=1}^{n} \beta^e \Delta S_t\ PDUM + \varepsilon_t,$$

here, β^m is the market risk, R_{mt} is the return on the country-specific market portfolio, β^e is the currency exposure of the portfolio, ΔS_t is the rate of change real exchange rates, $PDUM$ is the dummy variable for each of the three subperiods (1973–1980, 1981–1988, and 1989–1995), r_t is the monthly return, and is the error term. For the results, $ is the U.S. dollar, DM is the Deutschmark, and ¥ is the Japanese yen. Heteroscedastic-consistent t-statistics are in arentheses, and ***,**, * denotes 1%, 5%, and 10% significance levels, respectively. The adjusted R^2 shown is from the ordinary least squares regression ith an F-test of the variation across time shown in brackets. Firm variation is an F-test of the difference in exposure across firms.

		First period		Second period		Third period		
irm	Country-specific market risk	¥/$ Dummy	DM/$ Dummy	¥/$ Dummy	DM/$ Dummy	¥/$ Dummy	DM/$ Dummy	Adjusted R^2 (%) [Yen] [DM]
.S. portfolio	1.0619 (13.584)***	− 0.0393 (− 0.165)	0.0733 (0.260)	− 0.6673 (− 1.482)	0.3138 (1.063)	− 0.4184 (− 1.690)*	0.8987 (3.017)***	39.6 [1.125] [2.543]*
M	0.9105 (11.365)***	0.0918 (0.158)	0.0518 (0.204)	− 0.7549 (− 2.116)**	0.8363 (2.742)***	− 0.6313 (− 2.113)**	0.9836 (2.740)***	34.5 [2.610]* [3.816]**
ord	1.0254 (9.847)***	0.1935 (0.647)	− 0.0789 (− 0.279)	− 0.8481 (− 2.207)**	0.3211 (0.965)	− 0.2856 (− 0.927)	0.4894 (1.623)	33.6 [2.488]* [1.087]
hrysler	1.2527 (9.483)***	− 0.4114 (− 0.999)	0.2518 (0.589)	− 0.3933 (− 0.473)	− 0.2148 (− 0.438)	− 0.3432 (− 0.908)	1.2277 (2.836)***	25.1 [0.006] [2.306]*
irm variation [F-test]		[1.195]	[0.514]	[0.396]	[2.521]*	[0.784]	[2.878]*	

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- For the U.S. portfolio, the third-period yen exposure is -0.4184 and significant at the 10% level; the third-period Deutschmark exposure is 0.8987 and significant at the 1% level.
- GM has strong yen and Deutschmark exposures in the second and third periods.
- Table 6 shows that Japanese firms have strong dollar exposure in the second period, when Japanese sales in North America rise sharply.
- In the third period, Japanese dollar exposure becomes weaker, consistent with increased U.S. production acting as an operational hedge.
- Japanese exposure to the Deutschmark becomes more important in the third period as competition extends into Europe.

Economic message: Exposure is not constant. It changes with the evolution from national competition to international competition and then global production and sales.

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able 6
ariation of Japanese portfolio and firm-specific exchange rate exposure

egressions of the automotive industry for the total sample and three subperiods, using the following model:

$$r_t = \beta^m R_{mt} + \sum_{i=1}^{n} \beta^e \Delta S_t\ PDUM + \varepsilon_t,$$

here, β^m is the market risk, R_{mt} is the return on the country-specific market portfolio, β^e is the currency exposure of the portfolio, ΔS_t is the rate of change real exchange rates, $PDUM$ is the dummy variable for each of the three subperiods (1973–1980, 1981–1988, and 1989–1995), r_t is the monthly return, and is the error term. For the results, $ is the U.S. dollar, DM is the Deutschmark, and ¥ is the Japanese yen. Heteroscedastic-consistent t-statistics are in arentheses, and ***,**, * denotes 1%, 5%, and 10% significance levels, respectively. The adjusted R^2 shown is from the ordinary least squares regression ith an F-test of the variation across time shown in brackets. Firm variation is an F-test of the difference in exposure across firms.

		First period		Second period		Third period		
irm	Country-specific market risk	$/¥ Dummy	DM/¥ Dummy	$/¥ Dummy	DM/¥ Dummy	$/¥ Dummy	DM/¥ Dummy	Adjusted R^2 (%) [$] [DM]
panese portfolio	1.0079 (13.073)***	0.1858 (0.659)	− 0.3454 (− 1.541)	− 0.6538 (− 2.573)***	0.3938 (0.996)	− 0.0240 (− 0.190)	− 0.3032 (− 1.877)*	46.5 [4.234]** [2.428]*
oyota	0.8489 (7.385)***	− 0.0554 (− 0.222)	− 0.1347 (− 0.463)	− 0.8374 (− 2.557)**	0.4599 (0.806)	− 0.1205 (− 0.543)	− 0.1888 (− 0.843)	19.5 [1.665] [0.722]
lissan	0.8641 (9.859)***	− 0.1051 (− 0.411)	− 0.0548 (− 0.194)	− 0.4443 (− 1.747)*	0.2272 (0.524)	− 0.0998 (− 0.541)	− 0.3566 (1.448)	28.7 [0.560] [0.812]
londa	0.8830 (7.231)***	0.5832 (0.995)	− 1.0145 (− 2.337)**	− 1.3107 (− 3.623)***	0.3750 (0.660)	− 0.6513 (− 2.836)***	− 0.2021 (− 0.603)	24.8 [7.566]*** [3.100]**

(a) Table 6: Japanese exposure, part 1

	1.4409 (7.587)***	0.5188 (1.132)	0.2224 (0.624)	− 0.8787 (− 2.161)**	1.0058 (1.647)*	0.8964 (1.825)*	− 0.7018 (− 1.749)*	31.4 [4.700]*** [2.697]*
a	0.9901 (8.022)***	0.3569 (0.873)	− 0.6021 (− 1.565)	− 0.5704 (− 1.821)*	0.3388 (0.790)	− 0.4131 (− 1.365)	− 0.2918 (− 1.201)	25.2 [2.071] [1.402]
i	0.8424 (7.365)***	0.2499 (0.274)	− 0.6426 (− 1.122)	− 0.6394 (− 2.111)**	0.6181 (1.022)	− 0.1551 (− 0.641)	0.1266 (0.470)	15.7 [1.329] [2.075]
bishi	1.1432 (9.593)***	− 0.2201 (− 0.789)	− 0.2037 (− 0.601)	0.0842 (0.222)	− 0.2464 (− 0.541)	0.3766 (1.765)*	− 0.5034 (− 2.046)**	34.4 [0.816] [0.2308]
variation [F-test]		[1.344]	[2.382]**	[2.551]**	[0.735]	[1.944]*	[0.790]	

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(b) Table 6: Japanese exposure, part 2

6.6 Tables 7 and 8: foreign operations, foreign sales, and operational hedging

Read:

- Table 7 documents the expansion of foreign production and affiliations after 1970.
- U.S. firms have long-standing European operations, such as GM's Opel and Ford's European operations.
- Japanese firms expand production in North America during the 1980s and early 1990s, including Honda, Toyota, Nissan, Mazda, Mitsubishi, and Suzuki.
- Table 8 shows that U.S. sales as a share of global sales significantly increases Japanese firms' dollar exposure.
- For the Japanese portfolio, the U.S.-sales coefficient is -2.5832 with a t-statistic of -3.198.
- Toyota and Nissan show significant negative U.S.-sales coefficients.
- U.S. production has a positive coefficient for several firms, consistent with exposure reduction, although the portfolio coefficient is not statistically significant.

Economic message: Foreign sales create exposure, while foreign production can hedge it. The table helps explain why Japanese dollar exposure weakens after firms build production capacity in North America.

Table 7
Major foreign production, operations, and affiliations of U.S. and Japanese automotive firms since 1970, by year of initiation of operations.

Year	Company	Activity
Before 1970	Ford	Ford begins sales in Europe prior to WWII primarily in the U.K; exact date is unclear
	General Motors	GM buys Adam Opel of Germany in 1929 to begin its European operations
1971	Isuzu/GM	Affiliation agreement is signed by GM and GM purchases 24.2% of Isuzu
	Mitsubishi/Chrysler	Tie-up between Chrysler Corporation and Mitsubishi Corporation is announced
	Chrysler	Retains majority ownership in Chrysler de Mexico
1978	Honda	Honda of America Manufacturing, Inc. is established in Ohio, U.S.A.
1979	Ford/Mazda	Ford acquires 25% of Mazda
1980	Nissan	Nissan acquires shares in Motor Ibérica, S.A. in Spain
1981	Suzuki/GM	Tie-ups with General Motors Corporation of the U.S. and with Isuzu Motors Ltd. are concluded
1982	Suzuki	The first Suzuki passenger car comes off the line at Pakistan plant Production and sales contract for automobiles is formally concluded with Maruti Udyog LTD of India
1983	Nissan	Production of the Safari (Patrol) starts at Motor Ibérica, S.A. in Spain First Nissan truck rolls of the line in Tennessee
	Suzuki	The first Suzuki car comes off the line at Maruti Udyog LTD of India
1984	Honda	Automobile manufacturing company is established in Canada
	Nissan	Nissan begins sales of the Santana produced through cooperative agreement with Volkswagen
	Toyota/GM	Toyota–GM joint venture (New United Motor Manufacturing, Inc.) in U.S. starts production
	Chrysler	Chrysler purchases 16% of Maserati of Italy
1985	Mazda	Mazda Motor Manufacturing (USA) Corp. starts the construction of its Michigan plant
	Nissan	The Tennessee, USA plant begins production of the Sunny (Sentra) passenger car

(a) Table 7: foreign operations, part 1

Table 7 (*continued*)

Year	Company	Activity
1986	Honda	Production of Accords commences in Canada
	Mitsubishi/Chrysler	The plant of Diamond-Star Motors, an Mitsubishi Motors Corporation and Chrysler joint undertaking, is established
	Nissan	NMUK's vehicle assembly plant in North East England commences production of the Bluebird Arizona Test Center, Inc. (ITC) is established to bring the Company into closer contact with the world maker and begins operations
1987	Honda	Engine production commences in the U.S.
	Mazda	Vehicle production begins in U.S. plant
1988	Mitsubishi/Chrysler	Diamond-Star starts production of the new sports coupe
	Nissan	Nissan European Technology Centre Ltd. (NETC) in the U.K. begins to strengthen Nissan's research and development activities in Europe
	Toyota	Toyota Motor Manufacturing, U.S.A., Inc. starts production
		Toyota Motor Manufacturing Canada Inc. starts production
1989	Isuzu	New production plant in the U.S. for Isuzu and Subaru
	Nissan	Nissan Mexicana, S.A. de C.V. is established
	Chrysler	Joint venture begins with Steyr-Daimler-Puch of Austria to produce minivans and Jeeps for Europe
	Ford	Ford acquires Jaguar of the U.K.
	GM	GM purchases 50% of Saab of Sweden
1990	Suzuki	Suzuki reaches the basic agreement for production of passenger cars in Hungary by way of a joint venture with C. Itoh & Co., Ltd. and Autokonszern RT
1991	Mitsubishi	Mitsubishi Motors, the Dutch State, and Volvo Car Corporation sign an agreement on joint car production in The Netherlands
1992	Honda	Production of Accord begins in the U.K.
	Nissan/Ford	The Quest multipurpose vehicle, developed in a joint program with Ford Motor Co. of the U.S., begins production at a Ford plant in Ohio
	Suzuki	Automobile production starts in Hungary
	Toyota	Toyota Motor Manufacturing (U.K.) Ltd. starts production

(b) Table 7: foreign operations, part 2

Table 8
Foreign sales, production, and exchange rate interaction for Japanese firms

Regressions are shown of the Japanese firms for the full sample, using the following model:

$$r_t = \alpha + \beta^{m} R_{mt} + \gamma_1 S_{¥,t}\, TS_{JP} + \lambda_2 S_{¥,t}\, Pr_{JP,US} + \varepsilon_t,$$

where, β^{m} is the market risk, R_{mt} is the return on the country-specific market portfolio, Pr is the U.S. production as a percentage of U.S. sales, TS is the sales in the U.S. as a percentage of total global sales, S_t is the change in real exchange rates, r_t is the monthly return, and ε_t is the error term. Heteroscedastic-consistent t-statistics are in parentheses, and ***,**, * denotes 1%, 5%, and 10% significance levels, respectively.

Firm	Intercept	Country-specific market risk	U.S. sales/ total sales	U.S. production/ U.S. sales	Adjusted R^2 (%)
Japanese portfolio	0.0058	0.8040	− 2.5832	0.6253	52.2
(n = 154)	(1.657)*	(10.533)***	(− 3.198)***	(1.363)	
Toyota	0.0095	0.6532	− 5.9955	2.8295	25.5
(n = 109)	(1.442)	(4.530)***	(− 2.601)***	(2.223)**	
Nissan	0.0055	0.6924	− 2.6039	0.4734	33.3
(n = 127)	(1.040)	(7.364)***	(− 2.346)**	(0.681)	
Honda	0.0091	0.6000	− 1.3228	− 0.7832	20.3
(n = 154)	(1.445)	(4.671)***	(− 1.069)	(− 0.711)	
Isuzu	0.0206	2.0067	− 1.5506	1.0346	57.8
(n = 70)	(1.586)	(6.539)***	(− 0.399)	(0.482)	
Mazda	0.0094	0.9372	− 2.8415	0.3245	39.1
(n = 98)	(1.288)	(6.871)***	(− 1.147)	(0.207)	
Mitsubishi	0.0074	0.9208	− 7.9494	2.0001	57.7
(n = 70)	(1.222)	(6.827)***	(− 1.481)	(1.813)*	
Suzuki	0.0129	0.7575	18.4532	− 0.9972	37.8
(n = 70)	(1.696)*	(5.582)***	(1.011)	(− 1.394)	

Table 8: U.S. sales, U.S. production, and exchange-rate exposure for Japanese firms

7 Short summary

- The paper studies real exchange-rate exposure in the U.S. and Japanese automotive industry from 1973 to 1995.
- The automotive industry is chosen because it is globally competitive, export-intensive, and structurally changing.
- Exchange-rate exposure is modeled as the sensitivity of firm value to real exchange-rate movements.
- The theory emphasizes foreign sales, demand elasticity, cross elasticities, competition, production location, and hedging.
- The paper uses bilateral real exchange rates rather than trade-weighted indices.
- The U.S. auto portfolio has significant yen and Deutschmark exposure.
- Japanese firms have significant dollar exposure, especially Toyota, Nissan, and Honda.
- Market shares explain exposure: U.S. firms' yen exposure is tied to Japanese firms' U.S. market share.
- Exposures vary over time as the industry becomes more global and competitive.
- Japanese dollar exposure is strongest when U.S. sales rise but before U.S. production provides a substantial natural hedge.
- Foreign sales increase exposure; foreign production reduces exposure by matching revenues and costs in the same currency.
- The paper's main methodological lesson is that industry structure must be modeled when studying exchange-rate exposure.

8 Conclusion

8.1 Core conclusion

Williamson shows that exchange-rate exposure is significant in the automotive industry once the empirical design accounts for the relevant currencies, product-market competition, and operational structure. The paper's evidence supports the theoretical prediction that exposure is stronger in globally competitive industries with high substitutability and foreign sales.

8.2 Economic intuition

Currency movements change the relative costs and revenues of firms headquartered in different countries. In a monopolistic setting, firms may pass through exchange-rate changes and show little exposure. In a competitive setting, pass-through is limited, and exchange-rate movements shift market share and profit. If firms produce abroad, local costs hedge local revenues and reduce net exposure.

8.3 Takeaway

The paper's enduring message is that currency exposure is not a simple foreign-sales ratio. It is a joint outcome of foreign revenues, foreign costs, competitor location, market share, substitutability, and hedging. For empirical work, this means that exchange-rate exposure should be studied at the level of specific industries, currencies, and competitive relationships.